

Electric vehicles**Will a 3-minute battery swap beat a 5-minute charge for China's EV drivers?**

China's CATL to build thousands of swapping stations to speed up adoption of alternative charging technology



A Nio battery swapping station in China. The company is promoting 3-minute swapping times © FT montage/Getty Images

Edward White in Shanghai and **Gloria Li** in Hong Kong

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At a Nio battery-swapping station near Shanghai's Huangpu river, it takes only a brief voice command to start an automated process that will power up the electric vehicle in just three minutes.

Steering itself, the car halts above a retractable metal floor, before robotic arms remove its undercarriage and swap its depleted battery for a fully charged one. With a few hardware and software checks, the vehicle is ready for the road — and the next can swing into the station.

Nio, a US-listed Chinese EV start-up, runs more than 3,000 such stations in China. Its embrace of the technology has made it an outlier among its peers because most other manufacturers rely on recharging technology.

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An electric car using a Nio battery-swapping station in China © Bloomberg/Nio

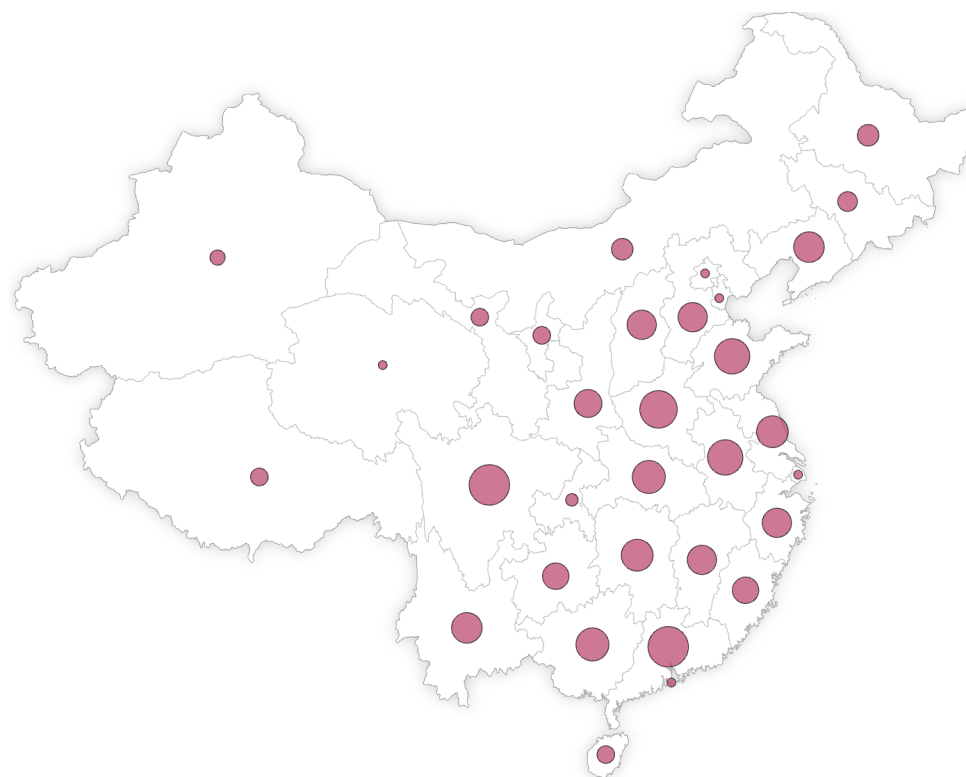
But a series of investments in swapping infrastructure by CATL, the world's largest battery maker, is now set to pave the way towards wider adoption in the world's biggest EV market.

CATL plans to build 1,000 swapping stations for passenger vehicles this year in China before expanding the network to about 10,000 by 2028 to serve 1mn cars each day, according to the company. It is also investing in battery-swapping systems for heavy trucks.

Duo Fu, a Shanghai-based EV sector analyst at consultancy Rystad Energy, said CATL's "phenomenal" technical capabilities and huge influence over the EV market meant the path ahead for swapping appears "much easier".

Nio's operational swapping stations across 285 Chinese cities

Number of cities with Nio vehicle swapping locations, by province



FINANCIAL TIMES

Source: Nio

Fu added that swap stations could take advantage of "peaks and valleys" in electricity pricing, whereas fast-charging technology — allowing a car to be fully charged within as little as five minutes from a fixed point — will require expensive electricity transmission upgrades before it becomes widely available.

"The conclusion we have reached is these two technologies . . . will not cannibalise each other," said Fu.

This year, EV sales are expected to surpass those of petrol-powered cars in China for the first time. To support the transition, the central government is expanding and upgrading both swapping and flash charging infrastructure.

In July, the National Development and Reform Commission, China's state planner, announced it would construct 100,000 fast charging stations by 2027. The Shanghai government has since March offered companies subsidies to cover up to 40 per cent of their investments into swapping stations.

Swapping proponents say battery-swapping technology is faster than charging and is a better fit for China's densely populated cities, where most people live in apartment buildings without their own charge point. Swapping can also **reduce the cost of an electric car by thousands of dollars**, as drivers do not own the batteries.

Alongside EV leader BYD, CATL has been a **pioneer in flash charging** as the companies race to ease driver fears over long charge times and EV ranges. However, the company's swapping investments are also a sign that it believes both technologies can be complementary as part of China's rapid EV transition.

Yang Jun, who leads CATL's swapping division, has said that as the EV industry moves into "uncharted waters" more standardised swapping equipment would benefit EV makers. The company estimates that 30,000 battery swap stations would be needed nationwide, compared with China's 100,000 petrol stations.

CATL said this month it would work with rental group CAR and lessor CMB Financial Leasing to build a fleet of 100,000 rental vehicles with swappable batteries.

Still, critics have doubts about the interoperability between different cars and batteries and point to the high cost of building swapping infrastructure.

He Xiaopeng, co-founder and chief executive of EV maker Xpeng, told the Financial Times that his company considered the technology "for five or six years" before "ultimately" deciding against it in about 2023.

"Advancing battery technology is [more important] than developing battery-swapping capabilities. That's the path we've chosen."

China's EV infrastructure keeps pace with sector growth as others fall behind

Expected change in light-duty vehicle charging points from 2024 to 2030 (mn)

Expected change in EVs per public charging points from 2024 to 2030

FINANCIAL TIMES

Source: IEA

CATL is also building 300 swap stations for heavy trucks this year, the early stages of a network it is planning with state-owned oil company Sinopec to eventually cover 150,000km across critical freight routes.

Robin Zeng, CATL's founder and chair, [told the FT in May](#) that he expected half of all new trucks sold in China to be electric by 2028. CATL has told the FT it plans to build swapping networks in Europe and other regions outside of China.

Christopher Doleman, an analyst with the Institute for Energy Economics and Financial Analysis, a US think-tank, said swapping was now part of a rapid transition to electric trucks, displacing both diesel and LNG-fuelled vehicles in China.

Doleman said that while electric heavy trucks could cost more up front than diesel counterparts, they become cheaper **over their lifespan as a result of lower fuel costs**.

Yuqian Ding, who leads HSBC's China automotive analysis, said CATL's embrace of battery swapping technology suggested it was set to become an "important" part of the broader electrification ecosystem.

"Scale is really the key," she said, adding that the swapping infrastructure will make CATL "like a new gas station" owner.

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